

2-Mark Questions & Answers -- MPCA Module 4: Memory & Peripherals Interfacing

Q1. What is the role of 74LS138 in memory interfacing

The 74LS138 is a 3-to-8 line decoder that translates 3 higher address bus lines into 8 active-low Chip Select (/CS) signals to uniquely enable individual RAM/EPROM memory chips.

Q2. What are the three operating modes of 8255 PPI

1. Mode 0 (Basic I/O): Simple input/output without handshaking.

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Q3. Define DMA and list the 4 channels of 8257 DMAC.

DMA (Direct Memory Access) is a high-speed data transfer mechanism directly between I/O peripherals and RAM without CPU intervention.

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Q4. Differentiate between IRR, ISR, and IMR in 8259 PIC.

- * IRR (Interrupt Request Register): Stores pending interrupt requests.

* ISF

Q5. What is absolute vs partial address decoding

- * Absolute Decoding: All address lines are decoded; no shadow addresses exist.

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